

Functional Test - Module 8: Edit User Profile

Project: Blue Team — Moffat Bay Marina

Course: CSD 460 - Capstone Project

Description: Test plan for the Edit User Profile page (Miguel — Front End, Robert — Back End).

Date: 2026/09/13

Front End Developer: Miguel Fernandez

Back End Developer: Robert Breutzmann

Front End Developer Tester: Miguel Fernandez

Back End Developer Tester: Robert Breutzmann

Front End Peer Tester: Sara White

Back End Peer Tester: Carolina Rodriguez

To complete the test case plan, fill out the information for Project, Course, Description and Date in the header above.

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For each test, the developer should: provide a test description, a test objective, the developer name and date tested. For each step, fill out the actions to be taken and describe the expected results, and enter **Pass** or **Fail**. The peer tester should provide their name, date tested, enter **Pass** or **Fail** for each step, and add each step's screenshot(s) in the same cell as their Pass or Fail.

Shared test account across Tests 1-4

All four of Robert's tests below run against one continuous test customer rather than the shared Elena Marsh demo account, so nothing here disturbs the account other testers rely on. Run them in order — 1, then 2, then 3, then 4 — since each one starts from the state the previous one left behind (the email changes in Test 1, three fields change in Test 2, the password changes in Test 3, and it changes again in Test 4). Starting test data, before Test 1's first step:

- First name: QA · Last name: EditTest
- Email: qa.edittest@example.com
- Password: TestPass1!
- Phone country code: 1 · Phone: 3605550199
- Street address: 100 Test Way · Address line 2: *(blank)*
- City: Anacortes · State: WA · ZIP: 98221 · Country: US

Separate test account for Tests 5-6

Miguel's two front-end tests use their own customer, created in Test 5's first step, so they can be run in any order relative to Robert's and neither set disturbs the other. Run Test 5 before Test 6, since Test 6 starts from the account Test 5 creates. Starting test data:

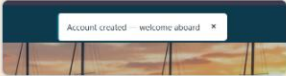
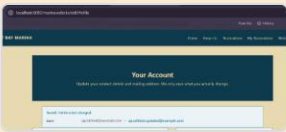
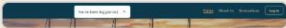
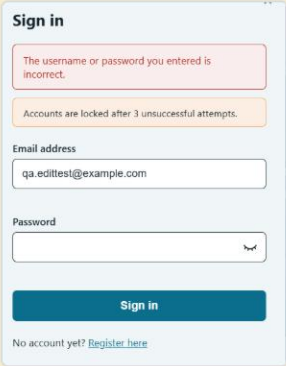
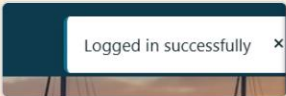
- First name: QA · Last name: Frontend
- Email: qa.frontend@example.com
- Password: TestPass1! (left as Newpass99! after Test 6)
- Phone country code: 1 · Phone: 3605550150
- Street address: 1 Test Way · Address line 2: *(blank)*
- City: Moffat Bay · State: WA · ZIP: 98110 · Country: US
- No boat registered

Test 1: Email Change Invalidates the Old Login

Test Objective: Verify that changing a customer's email address on Edit User Info takes effect for authentication immediately — the old email can no longer sign in, and only the new one can.

DEVELOPER: Robert Breutzmann | **TESTED ON:** Sept 16, 2026

PEER TESTER: Carolina Rodriguez | **TESTED ON:** Sept 20, 2026

Step	Action	Expected Results	Dev Test	Peer Test
1	Register a new customer at /register using the test data above (email qa.edittest@example.com , password TestPass1!).	Account is created and the page confirms success (registered notice / redirect to sign in).	Pass	Pass The registration success notice / redirect after Step 1's account creation. 
2	Sign in with qa.edittest@example.com / TestPass1! .	Login succeeds; header greeting reads "Welcome, QA E."	Pass	Pass Header greeting reading "Welcome, QA E." after Step 2's login. 
3	Click the header greeting to open Edit User Info. Change only the Email field to qa.edittest.updated@example.com , leave every other field as-is, click Save changes , confirm on the inline "Save these changes?" panel, then click Yes, save .	Page redirects to /editProfile?notice=profileUpdated ; a "Profile updated" toast appears; the "Saved. Here's what changed" summary lists exactly one row: Email, qa.edittest@example.com → qa.edittest.updated@example.com .	Pass	Pass The "Saved. Here's what changed" summary showing the Email row after Step 3. 
4	Log out via the header's Log Out button.	Redirected to the landing page; header shows Sign In again.	Pass	Pass The landing page header showing Sign In again after Step 4's logout. 
5	Attempt to sign in with the old email qa.edittest@example.com and password TestPass1! .	Login fails with the generic "email or password is incorrect" message (the same anti-enumeration message used for any bad credential, not an account-not-found message).	Pass	Pass The login modal's generic incorrect-credentials error after Step 5's old-email attempt. 
6	Sign in with the new email qa.edittest.updated@example.com and password TestPass1! .	Login succeeds; header greeting reads "Welcome, QA E." again.	Pass	Pass Header greeting reading "Welcome, QA E." again after Step 6's new-email login. 

Comments

- Developer (Robert)**

I added a toggle for password visibility so you can see what you are typing.
- Developer (Robert)**

Something I noticed that is unrelated to this test is that when you create an account, you are not automatically logged in. Some sites do this where when you create an account you are logged in and directed to where you had attempted to reach. In the case of this test, that's back on the landing page. Worth looking at the registration contract to see if creating an account should log you in as that account automatically.
- Peer Tester (Carolina)**

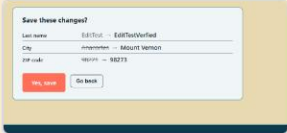
All six steps passed as expected. The email change saved correctly, the old email no longer worked for login, and the updated email successfully signed the user back in. I did not run into any issues during this test.

Test 2: Partial Save Writes Only the Changed Columns

Test Objective: Verify that saving a partial profile update writes exactly the fields the customer changed to the `Customer` table, and leaves every other column — including `passwordHash`, `dateJoined`, and the lockout columns — byte-for-byte unchanged.

DEVELOPER: Robert Breutzmann | **TESTED ON:** Sept 16, 2026

PEER TESTER: Carolina Rodriguez | **TESTED ON:** Sept 20, 2026

Step	Action	Expected Results	Dev Test	Peer Test
1	Signed in as <code>qa.edittest.updated@example.com</code> (continuing from Test 1), run the "Before" query below against the database.	Query returns exactly one row for this customer; record every column's value.	Pass	Pass The database client showing the "Before" row returned by Step 1's query. 
2	On Edit User Info, change exactly three fields — Last name to <code>EditTestVerified</code> , City to <code>Mount Vernon</code> , ZIP code to <code>98273</code> — and leave every other field untouched. Click Save changes , review the confirmation panel, then click Yes, save .	The confirmation panel (and, after saving, the "Saved. Here's what changed" summary) lists exactly three rows: Last name, City, ZIP code — no others.	Pass	Pass The "Saved. Here's what changed" summary after Step 2, showing exactly Last name, City, and ZIP code. 
3	Run the "After" query below against the database.	Query returns exactly one row for this customer.	Pass	Pass The database client showing the "After" row returned by Step 3's query. 
4	Diff the "Before" and "After" rows column by column.	<code>lastName</code> = <code>EditTestVerified</code> , <code>city</code> = <code>Mount Vernon</code> , <code>zipCode</code> = <code>98273</code> . Every other column — <code>firstName</code> , <code>email</code> , <code>phone</code> , <code>phoneCountryCode</code> , <code>streetAddress</code> , <code>streetAddress2</code> , <code>state</code> , <code>country</code> , <code>passwordHash</code> , <code>dateJoined</code> , <code>failedLoginAttempts</code> , <code>accountLocked</code> — is identical to the "Before" row.	Pass	Pass The Before and After rows shown together (side-by-side or stacked), with Step 4's confirmed matches/differences visible. 

Run this query before Step 2 and again after, same query both times:

```
SELECT customerID, firstName, lastName, email, phone, phoneCountryCode,
       streetAddress, streetAddress2, city, state, zipCode, country,
       passwordHash, dateJoined, failedLoginAttempts, accountLocked
FROM Customer
WHERE email = 'qa.edittest.updated@example.com';
```

Expected row count: 1, both times. Expected diff: only `lastName`, `city`, `zipCode` change; all twelve other columns match between the two runs.

Comments

Developer (Robert)
The test successfully updated as expected. I ran into no issues with the process.

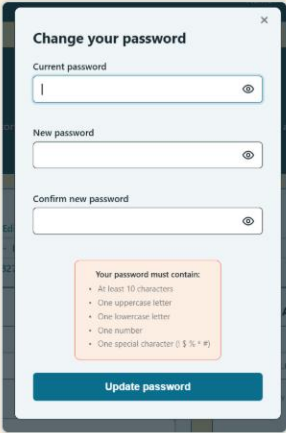
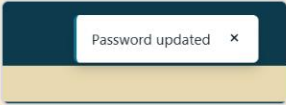
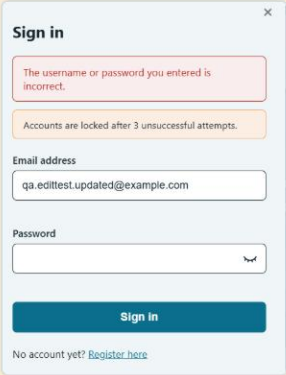
Peer Tester (Carolina)
All four steps passed as expected. The profile update only changed the Last Name, City, and ZIP code, and the before/after database check confirmed the remaining customer fields stayed the same. I did not run into any issues during this test.

Test 3: Change Password Retires the Old One Immediately

Test Objective: Verify that the Change Password modal on Edit User Info updates the stored password hash correctly, and that only the new password authenticates afterward.

DEVELOPER: Robert Breutzmann | **TESTED ON:** Sept 16, 2026

PEER TESTER: Carolina Rodriguez | **TESTED ON:** Sept 20, 2026

Step	Action	Expected Results	Dev Test	Peer Test
1	Signed in as qa.edittest.updated@example.com / TestPass1! (continuing from Test 2), open Edit User Info and click Change password .	The Change Password modal opens with Current password, New password, and Confirm new password fields, plus the live password rules checklist.	Pass	Pass The Change Password modal open, mid-fill, with the rules checklist visible (Step 1). 
2	Enter Current password TestPass1! , New password TestPass2! , Confirm new password TestPass2! , and click Update password .	The modal closes with no page navigation; a "Password updated" toast appears.	Pass	Pass The "Password updated" toast after Step 2. 
3	Log out via the header's Log Out button.	Redirected to the landing page; header shows Sign In again.	Pass	Pass The landing page header showing Sign In again after Step 3's logout. 
4	Attempt to sign in with qa.edittest.updated@example.com and the old password TestPass1! .	Login fails with the generic incorrect-credentials message.	Pass	Pass The login modal's incorrect-credentials error after Step 4's old-password attempt. 
5	Sign in with qa.edittest.updated@example.com and the new password TestPass2! .	Login succeeds; header greeting reads "Welcome, QA E."	Pass	Pass Header greeting reading "Welcome, QA E." after Step 5's new-password login. 

Comments

Developer (Robert)
I ran into no issues with this test. Everything functioned as expected and no unexpected issues came up. Same note from above about the password error and username error presenting the same results.

Peer Tester (Carolina)
Everything worked as expected during testing. The password change went through, the old password was rejected, and the new password worked for login. No issues came up during this test.

Test 4: Lockout Reset Clears the Lock and Replaces the Password

Test Objective: Verify that three failed sign-in attempts lock the account, that the forgot-password reset (reached from the login modal's locked-out state) both clears the lockout and sets a new password, and that only the new password works afterward.

DEVELOPER: Robert Breutzmann | **TESTED ON:** Sept 16, 2026

PEER TESTER: Carolina Rodriguez | **TESTED ON:** Sept 20, 2026

Step	Action	Expected Results	Dev Test	Peer Test
1	Attempt to sign in as qa.edittest.updated@example.com with an incorrect password three times in a row (e.g. wrong1 , wrong2 , wrong3).	The first two attempts show the generic incorrect-credentials message. After the third, the login modal switches to its locked-out state: the sign-in form is replaced with a message and a Reset your password button.	Pass	Pass The login modal's locked-out state after Step 1's third failed attempt, showing the Reset your password button. Sign in The username or password you entered is incorrect. Accounts are locked after 3 unsuccessful attempts. Email address qa.edittest.updated@example.com Password Sign in No account yet? Register here Sign in This account has been locked after multiple failed login attempts. Unlocking an account means setting a new password. We'll send a verification code to the email on the account. Reset your password
2	Click Reset your password .	The Forgot Password modal opens, with the Email field pre-filled with qa.edittest.updated@example.com .	Pass	Pass The Forgot Password modal from Step 2, with the email pre-filled. Reset your password Enter your email and we'll send a verification code. Resetting your password also unlocks an account that's been locked. Email address qa.edittest.updated@example.com Verification code Demo site - the code is 12345. New password Confirm new password Your password must contain: - At least 10 characters - One uppercase letter - One lowercase letter - One number - One special character (! \$ % ^ &) Reset password

Comments

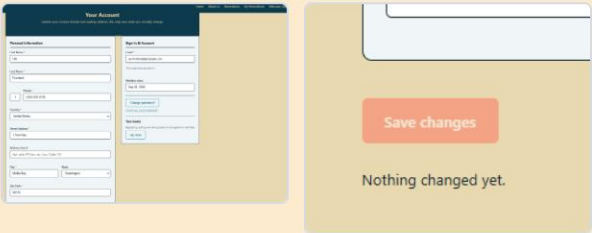
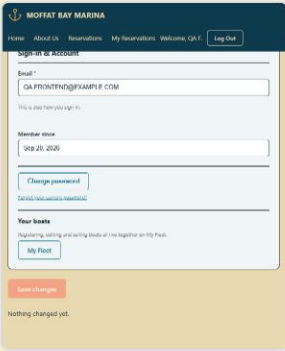
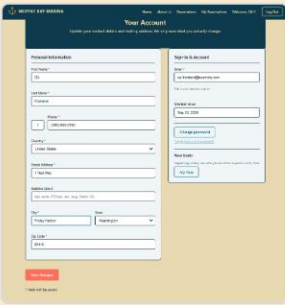
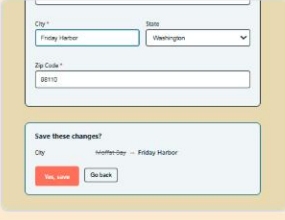
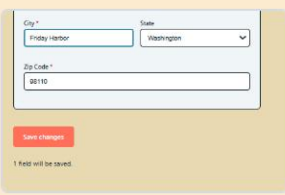
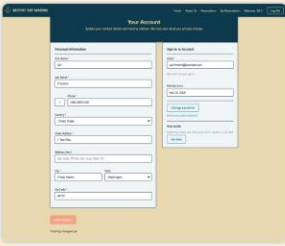
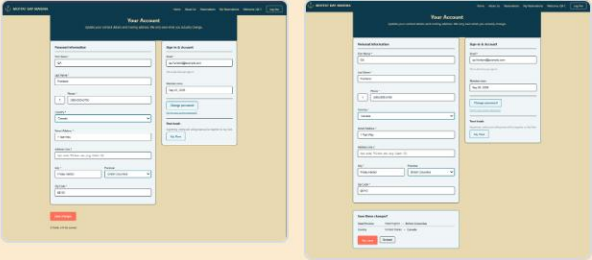
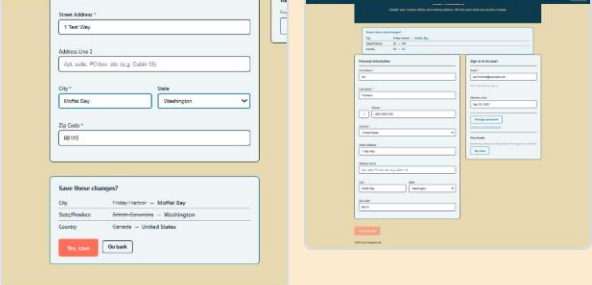
Developer (Robert) I ran into no issues with this test. Everything functioned as expected and no unexpected issues came up. Same note from above about the password error and username error presenting the same results.
Peer Tester (Carolina) Everything worked as expected during this test. The account locked after three failed attempts, the reset cleared the lockout, and the old password was rejected afterward. I was able to sign back in with the new password and did not run into any issues.

Test 5: Profile Form, Partial Update and Confirmation

Test Objective: Verify that the Edit User Profile form renders pre-filled from the signed-in customer's session, that only fields the customer actually changed are submitted, that the confirmation panel lists those changes accurately before anything is written, and that the post-save summary reports them afterwards and only once.

DEVELOPER: Miguel Fernandez | TESTED ON: Sept 19, 2026

PEER TESTER: Sara White | TESTED ON: September 20, 2026

Step	Action	Expected Results	Dev Test	Peer Test
1	Register a new customer at /registration.jsp : first name QA , last name Frontend , phone (360) 555-0150 , country United States, street 1 Test Way , city Moffat Bay , state Washington, ZIP 98110 , email qa.frontend@example.com , password TestPass1! . Leave the whole Boat Information column empty.	Account is created and the site confirms it with the "Account created — welcome aboard" popup.	Pass	Pass 
2	Sign in as qa.frontend@example.com / TestPass1! , then click the "Welcome, QA F." greeting in the header.	The account page loads at /editProfile with every field already filled in from the account — including the phone shown as (360)-555-0150 rather than as bare digits. "Save changes" is greyed out and the hint below it reads "Nothing changed yet."	Pass	Pass 
3	Retype the email in capitals as QA.FRONTEND@EXAMPLE.COM and move focus out of the field.	"Save changes" stays greyed out and the hint still reads "Nothing changed yet." Changing only the letter case is not a change, because the server lowercases the email before it compares — so the page compares the same way rather than offering to save something that would write nothing.	Pass	Pass 
4	Restore the email to qa.frontend@example.com , then change City from Moffat Bay to Friday Harbor .	The City field picks up a blue border marking it as edited, "Save changes" becomes clickable, and the hint changes to "1 field will be saved." No other field is marked.	Pass	Pass 
5	Click "Save changes".	The Save button is replaced by a panel headed "Save these changes?" listing exactly one row — City Moffat Bay → Friday Harbor , old value struck through — with "Yes, save" and "Go back" buttons. Nothing has been written to the database yet.	Pass	Pass 
6	Click "Go back".	The panel disappears and the Save button returns, still enabled. City still reads Friday Harbor — "Go back" returns the customer to editing, it does not undo the edit.	Pass	Pass 
7	Click "Save changes" again, then "Yes, save".	The page reloads at /editProfile , the "Profile updated" popup appears, and a summary box above the form reads "Saved. Here's what changed:" with the single row City Moffat Bay → Friday Harbor .	Pass	Pass 
8	Refresh the page.	The summary box is gone and no second save takes place. The form still shows Friday Harbor . The summary is shown once and does not re-announce an old edit on refresh.	Pass	Pass 
9	Change Country to Canada , choose British Columbia for Province, then click "Save changes".	As soon as Canada is chosen the State label becomes "Province" and the list becomes Canadian provinces. The confirmation panel then lists three rows using readable names rather than stored codes: City , State/Province Washington → British Columbia , and Country United States → Canada . Washington is still named even though it is no longer one of the options.	Pass	Pass 
10	Click "Yes, save".	All three changes save and the summary lists City, State/Province and Country. Note the difference in wording: the summary reports the stored values (WA → BC , US → CA) where the confirmation panel in Step 9 named them in full. Both are correct, but see the developer comment below.	Pass	Pass 
11	Set Country back to United States , State to Washington and City to Moffat Bay , then save.	The account returns to its starting values, and "Save changes" goes back to greyed out with "Nothing changed yet." — confirming the page recognises that nothing differs from what is on file.	Pass	Pass 

Database Check (Step 7)

```
USE MoffatBayMarinaDB;  
SELECT firstName, city, state, country FROM Customer WHERE email = 'qa.frontend@example.com';
```

Expected after Step 7: one row, QA | Friday Harbor | WA | US . firstName , state and country are untouched, which is what "only what changed is submitted" means in practice.

Comments

Developer (Miguel)
All eleven steps pass. Two worth understanding rather than just ticking. Step 3 looks like a non-event but it is the point of the partial update: the servlet lowercases the email before building its own diff, so the page has to compare the same way or it would offer to save a change that writes nothing. Step 9 needed a fix during development — the confirmation panel was reading whichever option was selected at that moment for both sides of the arrow, so it showed "British Columbia → British Columbia"; the label each field held when the page loaded is now captured on load, which is also why Washington can still be named after the province list has replaced it.

One thing I did *not* fix, for the peer tester to be aware of rather than to raise as a failure: Step 9 names values in full and Step 10 reports them as codes, one right after the other. The panel is built in the browser from the dropdown labels; the summary is rendered by the servlet from the stored column values, which have no idea "WA" means Washington. Both are accurate and nothing is wrong, but a customer reading them back to back has to work out they are the same thing. Fixing it properly means the servlet putting display labels in the diff, which is back end work — logging it for Module 9 polish.

Also for the peer tester: after Step 9 the State field is genuinely blank until a province is chosen, and saving at that point gives "Choose a state or province." under the field. That is correct behaviour, not a bug.

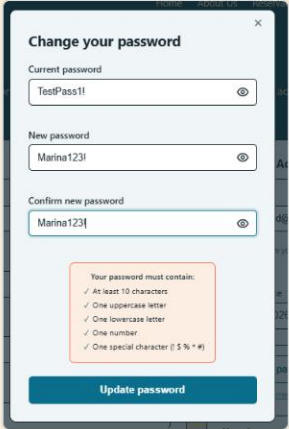
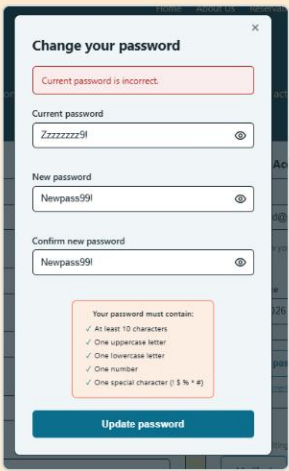
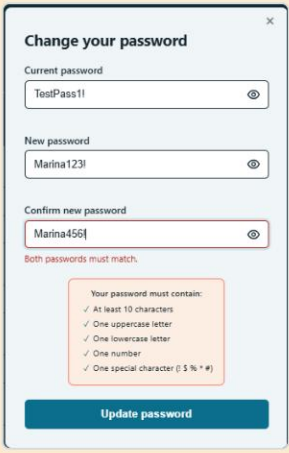
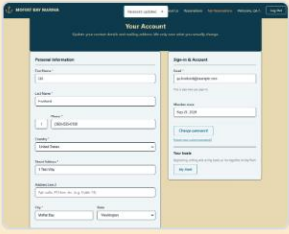
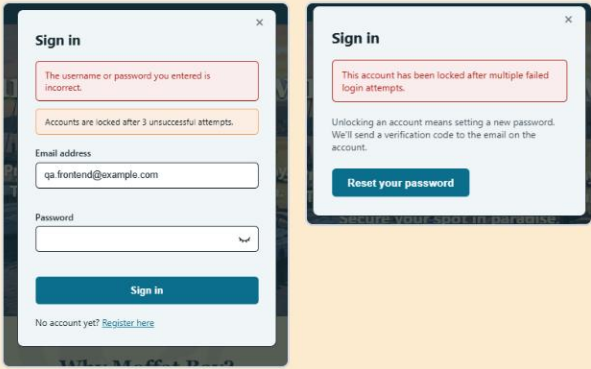
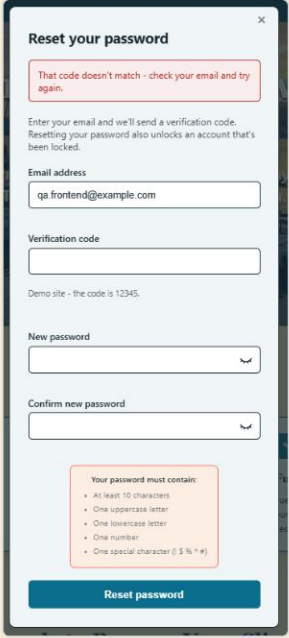
Peer Tester (Sara)
Everything worked as expected. I think it was a really good call to put in summaries of the changes before and after they are finalized - this gives the user plenty of opportunities to review the changes they're making and confirm the changes after they've taken place. For steps 7 and 10, the popups did appear, but I was unable to grab a screenshot of them before they disappeared. This is expected behavior since these types of confirmation popups typically disappear after a short time.

Test 6: Password Modals, Front End Behaviour

Test Objective: Verify the front end of both password flows — that the change-password modal refuses a wrong current password and a mismatched confirmation without leaving the page, and that the reset reached from the locked-out login modal never reveals whether an email address is registered.

DEVELOPER: Miguel Fernandez | TESTED ON: Sept 19, 2026

PEER TESTER: Sara White | TESTED ON: September 20, 2026

Step	Action	Expected Results	Dev Test	Peer Test
1	Signed in as qa.frontend@example.com / TestPass1! on /editProfile, click "Change password".	A modal opens headed "Change your password" with Current, New and Confirm fields, and the "Your password must contain" checklist inside it with all five rules unticked.	Pass	Pass 
2	Type Marina123! into New password and watch the checklist.	The rules tick off one at a time as each is satisfied, all five green by the end. The checklist is the modal's own, so Registration's identical checklist on another page is unaffected.	Pass	Pass 
3	Enter Current Zzzzzzz9! (deliberately wrong), New Newpass99!, Confirm Newpass99!, and click "Update password".	A red banner appears inside the modal reading "Current password is incorrect." The modal stays open, the page does not reload, and nothing is written — the current password is checked on the server, not just in the browser.	Pass	Pass 
4	Enter Current TestPass1!, New Marina123!, Confirm Marina456!, and click "Update password".	"Both passwords must match." appears under the Confirm field. Nothing is sent to the server — a typo is caught in the browser before it can become the customer's password.	Pass	Pass 
5	Correct Confirm to Marina123! and click "Update password".	The modal closes and a "Password updated" popup appears. The page does not reload — this flow answers with JSON rather than a page, unlike the reset below.	Pass	Pass 
6	Log out and sign in with qa.frontend@example.com / Marina123!.	Sign-in succeeds. The old password no longer works.	Pass	Pass 
7	Log out. Attempt to sign in three times as qa.frontend@example.com with the wrong password Wrongpass9!.	The first two attempts give the generic "The username or password you entered is incorrect." with the notice that accounts lock after 3 attempts. The third gives "This account has been locked after multiple failed login attempts." together with a "Reset your password" button — the only route back in, since a locked-out visitor has no session to reach an account page with.	Pass	Pass 
8	Click "Reset your password". In the modal that opens, enter code 99999 (wrong) with New and Confirm both Newpass99!, and submit.	The modal opens with qa.frontend@example.com already in the email field, carried over from the failed sign-in. On submit it re-renders already open with "That code doesn't match - check your email and try again." at the top.	Pass	Pass 
9	Change the email to nobody@example.com, leaving the code as 99999 (wrong) with New and Confirm both Newpass99!, and submit.	The message is word for word identical to Step 8, in the same place and the same styling. Nothing distinguishes an address that has an account from one that does not, so this endpoint cannot be used to find out which emails are registered.	Pass	Pass 
10	Set the email back to qa.frontend@example.com, enter code 12345, New and Confirm both Newpass99!, and submit.	The page redirects to the landing page and the "Password reset — sign in with your new password" popup appears.	Pass	Pass 
11	Sign in with qa.frontend@example.com / Newpass99!.	Sign-in succeeds on the first attempt, with no lockout message — the reset cleared the lock and set the new password together, rather than only changing the password.	Pass	Pass 